

Stony Brook University
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Spring 2018

ASSIGNMENT 3

Due: March 21, 2018, in class

INSTRUCTIONS: Show all your work to get full points. Please also cut and paste at the end of your submission the R code you have used for the last two problems.

1. Simple Regression in a Spreadsheet. Suppose you have the following data on 11 student's combined SAT scores X and their cumulative grade point average at graduation Y .¹

Y_i	X_i
3.63	1490
2.37	1300
3.33	1510
3.32	1420
3.27	1490
2.37	1180
3.61	1550
3.23	1460
2.59	1300
3.30	1450
3.21	1550

(a) Write down the econometric model relating X and Y (i.e. the linear regression model).

(b) Calculate the following:

$$n; \bar{X}; \bar{Y}; \sum (X_i - \bar{X})(Y_i - \bar{Y}); \sum (X_i - \bar{X})^2; \hat{\beta}_0 \text{ and } \hat{\beta}_1.$$

(c) Calculate $\hat{Y}_i$ and $\hat{u}_i$ for each observation.

(d) Calculate R^2 .

(e) Compute $\sum \hat{u}_i$. Explain why do you get this result.

2. Consider the savings function

$$sav = \beta_0 + \beta_1 inc + u, \quad u = \sqrt{inc} \cdot e,$$

where e is a random variable $E(e) = 0$ and $\text{Var}(e) = \sigma_e^2$. Assume that e is independent of inc .

¹Do not do this exercise by hand. Please do this in any Spreadsheet software that you are comfortable with. Another option would be to do it R. In any case you **should not use** any predefined command to calculate the OLS intercept and slopes.

- (a) Show that $E(u|inc) = 0$, so that the key *zero conditional mean assumption* is satisfied. *Hint: If e is independent of inc , then $E(e|inc) = E(e)$.*
- (b) Show that $Var(u|inc) = \sigma_e^2 \cdot inc$. *Hint: $Var(e|inc) = Var(e)$ if e and inc are independent.*
- (c) Provide a discussion that supports the assumption that the variance of savings increases with family income.

3. First Look at Regression Output. Suppose that a researcher, using data on class size (CS) and average test scores from 100 third-grade classes, estimates the OLS Regression

$$\widehat{TestScore} = 520.4 - 5.82 CS, \quad R^2 = 0.08, SER = 11.5,$$

(20.4) (2.21)

where the standard errors for the coefficient estimates are given by the numbers in parentheses.

- (a) Suppose a classroom has 20 students. What is the regression's prediction for that classroom's average test score?
- (b) Last year a classroom had 19 students, and this year it has 21 students. What is the regression's prediction for the change in the classroom average test score?
- (c) The sample average class size across the 100 classrooms is 21.4. What is the sample average of the test scores across the 100 classrooms? (*Hint: Review the formulas for the OLS estimators.*)
- (d) What is the sample standard deviation of the test scores across the 100 classrooms? (*Hint: Review the formulas for the R^2 and SER .*)
- (e) Construct a 95% confidence interval for β_1 , the regression slope coefficient.
- (f) Calculate the p -value for the two-sided test of the null hypothesis $H_0 : \beta_1 = 0$. Do you reject the null hypothesis at the 5% level? At the 1% level?
- (g) Calculate the p -value for the two-sided test of the null hypothesis $H_0 : \beta_1 = -5.6$. Without doing any additional calculations, determine whether -5.6 is contained in the 95% confidence interval for β_1 .
- (h) Construct a 99% confidence interval for β_0 .

4. Suppose that a researcher, using wage data on 250 randomly selected male workers and 280 female workers, estimates the OLS regression

$$\widehat{Wage} = 12.52 + 2.12 Male, \quad R^2 = 0.06, SER = 4.2,$$

(0.23) (0.36)

where $Wage$ is measured in dollars per hour and $Male$ is a binary variable that is equal to 1 if the person is male and 0 if the person is a female. Define the wage gender gap as the difference in mean earnings between men and women.

- (a) What is the estimated gender gap?
- (b) Is the estimated gender gap significantly different from 0? (Compute the p -value for testing the null hypothesis that there is no gender gap.)
- (c) Construct a 95% confidence interval for the gender gap.
- (d) In the sample, what is the mean wage of women? Of men?
- (e) Another researcher uses these same data but regresses $Wages$ on $Female$, a variable that is equal to 1 if the person is female and 0 if the person is a male. What are the regression estimates calculated from this regression?

5. Empirical Exercise: Simple Linear Regression. To solve this problem you need to use the same dataset as in the second assignment (i.e., CPS08) posted on Blackboard. The dataset contains information on wages, gender, education level, and type of employment from 500 employees. The goal of this exercise is to explore the relationship between earnings and age.

Assume that the four *OLS* assumptions discussed in class are satisfied. Estimate the following simple regression model of *LogWage* on *Age*

$$\text{LogWage}_i = \beta_0 + \beta_1 \text{Age}_i + u_i, \quad i = 1, \dots, N.$$

- (a) Report the estimates for $\hat{\beta}_0$ and $\hat{\beta}_1$. Interpret the estimate for the slope parameter.
- (b) Compute the R^2 for this regression? How can you interpret its value?
- (c) Repeat parts (a) and (b) but using the observations for female workers (i.e., $FEMALE = 1$).
- (d) Repeat parts (a) and (b) but using the observations for male workers (i.e., $FEMALE = 0$).

6. Empirical Exercise. Use the data in COUNTYMURDERS to answer these questions. Use only the data for 1996. The variable *murders* is the number of murders reported in the county. The variable *execs* is the number of executions that took place of people sentenced to death in a given county. Most states in the United States have the death penalty, but several do not.

- (a) How many counties are there in the data set? How many counties had zero murders in 1996? How many counties had at least one execution? What is the largest number of executions? Why is the average number of executions so small?
- (b) Estimate the equation

$$\text{murders} = \beta_0 + \beta_1 \text{execs} + u$$

by OLS and report the results in the usual way, including sample size and R^2 .

- (c) Interpret the slope coefficient in part (b). Does the estimated equation suggest a deterrent effect of capital punishment?
- (d) What is the smallest number of murders that can be predicted by the equation? What is the residual for a county with zero executions and zero murders?
- (e) Explain why a simple regression analysis is not well suited for determining whether capital punishment has a deterrent effect on murders.